

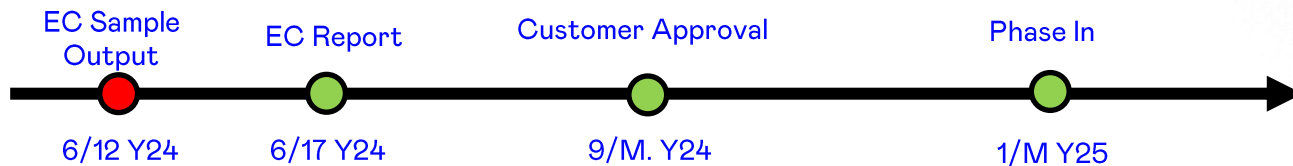
G230HAN01.1 Engineering Change Report

2024/3/26

Summary



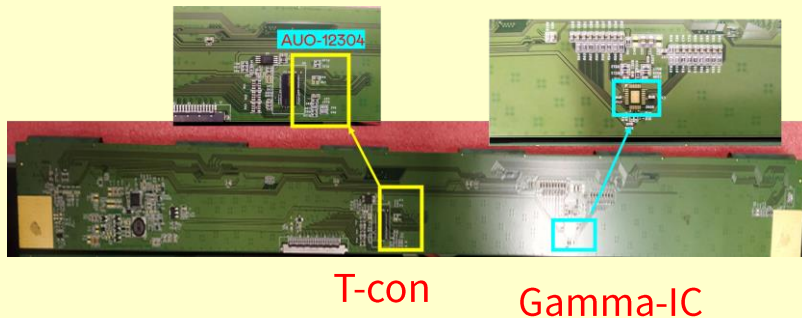
Schedule Plan:



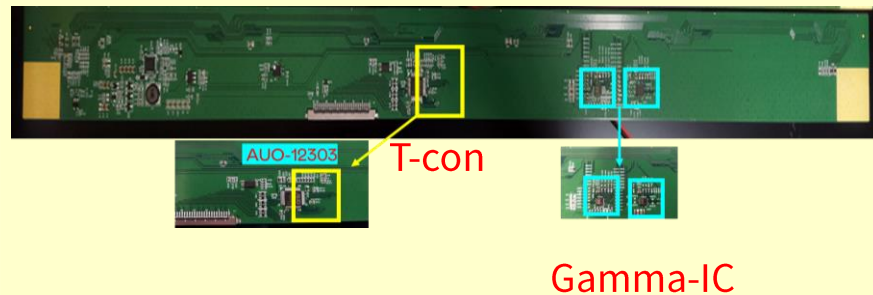
Item	EC item	Current	New	Background and Purpose
1	T-con & PCB re-layout	AUO-12304	AUO-12303	The current material is EOL. Due to T-con changed, PCB needed to be re-layout.
2	Vcom	VR knob	Vcom (Digital mode)	The current material Vcom EOL, and the VR knob no longer manufacturing; which current Vcom trend changed to digital mode already. Due to R-string is sharing the same Vcom, the Gamma IC changed to 18 groups of voltages.

PCBA comparison

Before



After



- Due to T-con /Gamma IC being changed, PCB needed to be re-layout. The PCBA appearance is different from the original.

Optical Report

Optical Measurement Report						
Originator	chun li		Model/PCS	G230HAN01.1/5PCS	Date	2024/2/12
Test Condition	Lum.,CR,Chromaticity,Uniformity,Gary Scale,V-T,View Angle,RT,CrossTalk					
Test System	Detector		Inverter	DC Voltage	Warm up time	
FPM-520-2	SR-UL1R		NA	12V	30min	

【Test Reason】

G230HAN01.102 New PCBA驗證

【Judgement】

PASS

Test Items	unit	PSS					H9FF7MBZZZS0	H9FF7RHZZZS0	H9FF7AGZZZS0	H9FF7MHZZZS0	H9FF7PAZZZS0	-4σ	mean	+4σ
							20013CQ000M108	20013CQ000M108	20013CQ000M108	20013CQ000M108	20013CQ000M108			
		min	typ	max										
Center Brightness	cd/m ²	240	300		341.1	348.0	346.8	349.4	348.0	333.8	346.7	359.6		
Darkness (L0)					0.24	0.25	0.24	0.25	0.25	0.239	0.25	0.253		
Contrast Ratio(CR)		600	1000		1393.1	1404.1	1421.3	1413.2	1404.9	1364.9	1407.3	1449.7		
Brightness(5P Ave.)	cd/m ²				329.8	335.9	333.8	334.9	331.8	323.3	333.2	343.1		
Chromaticity														
Wx	---	0.263	0.313	0.363	0.297	0.295	0.298	0.295	0.294	0.290	0.296	0.302		
Wy	---	0.279	0.329	0.379	0.326	0.320	0.322	0.319	0.319	0.309	0.321	0.333		
Rx	---	0.593	0.643	0.693	0.646	0.642	0.642	0.642	0.645	0.637	0.644	0.651		
Ry	---	0.285	0.335	0.385	0.335	0.335	0.335	0.335	0.335	0.334	0.335	0.336		
Gx	---	0.256	0.306	0.356	0.296	0.296	0.298	0.296	0.297	0.294	0.297	0.300		
Gy	---	0.557	0.607	0.657	0.608	0.608	0.608	0.609	0.607	0.605	0.608	0.611		
Bx	---	0.105	0.155	0.205	0.152	0.152	0.153	0.153	0.152	0.151	0.152	0.154		
By	---	0.012	0.062	0.112	0.064	0.063	0.062	0.062	0.061	0.058	0.062	0.067		
NTSC	---		71	---	72.6	72.1	72.0	72.3	72.5	71.195	72.308	73.422		
CCT	---		---	---	7499	7671	7457	7740	7773	7058.2	7627.9	8197.6		

- For the validation of G230HAN01.102 with EC T-con & Vcom: Optical Result PASS.

RA Result

Product No	G230HAN01.0	Part No	97.23G02.002	Applicant/Dept	仇栋琴/RRPBF3					Judgement	
Test Purpose	G230HAN01.002 UV328 free+New PCBA驗證			Application Date	2024/02/02						
				Input Date	2024/02/09						
RA No	D240207024	Product Stage	DVT	Issue Date							
Test Item	Test Condition	Test Panel			Test Result						
HTO	50°C 500h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	T500	Judgement	
		D2402070240001		ZD3H9FF7HFZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240002		ZD3H9FF7BHZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240003		ZD3H9FF7CHZZS0102	OK	OK	OK	OK	OK	PASS	
LTO	0°C 500h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	T500	Judgement	
		D2402070240004		ZD3H9FF7FAZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240005		ZD3H9FF7NFZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240006		ZD3H9FF7GBZZS0102	OK	OK	OK	OK	OK	PASS	
THB	50°C 80% 500h	No	Chip_Id	Shipping_No	T0★	T100★	T200★	T300★	T500	Judgement	
		D2402070240007		ZD3H9FF7JHZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240008		ZD3H9FF7NAZZS0102	OK	OK	OK	OK	OK	PASS	
		D2402070240009		ZD3H9FF7NCZZS0102	OK	OK	OK	OK	OK	PASS	
TST	-20°C (30min) ~ 60°C (30min), 100cycles judge, 200cycles reference, 200cycles	No	Chip_Id	Shipping_No	C0★	C100★	C200			Judgement	
		D2402070240010		ZD3H9FF7JDZZS0102	OK	OK	OK			PASS	
		D2402070240011		ZD3H9FF7TFZZS0102	OK	OK	OK			PASS	
		D2402070240012		ZD3H9FF7GDZZS0102	OK	OK	OK			PASS	

- The Polarizer validation of UV-328 free for G230HAN01.0 showed PASS result at T500 in RA.

IS result long time & short time : PASS

SN (至少2位)	关机16h后, 开机判				
	殘影類型	L255	L128	L64	L0
ZD3H9FF7PAZZZS0102	S	无	无	无	无
	L	无	无	无	无
ZD3H9FF7RHZZZS0102	S	无	无	无	无
	L	无	无	无	无
ZD3H9FF7MBZZZS0102	S	无	无	无	无
	L	无	无	无	无

Remark : G230HAN01.102 IS result for long time : **PASS**

SN (至少2位)	殘影類型	1.5hour-10min				
		L255	L127	L64	L32	L0
ZD4H1E28GDZZZS0102	S	无	无	无	无	无
	L	无	无	无	无	无
ZD4H1E65CHZZZS0102	S	无	无	无	无	无
	L	无	无	无	无	无
ZD4H1E28GEZZZS0102	S	无	无	无	无	无
	L	无	无	无	无	无

Remark : G230HAN01.102 IS result for short time : **PASS**

ESD result

(1) Contact Discharge:

Voltage	Sample1	Sample2	Sample3
	ZD3H9FF7NEZZ-ZS0102	ZD3H9FF7JGZZ-ZS0102	ZD3H9FF7GAZZ-ZS0102
+ 8 KV	B	B	B
- 8 KV	B	B	B

(2) Air Discharge:

Voltage	Sample1	Sample2	Sample3
	ZD3H9FF7NEZZ-ZS0102	ZD3H9FF7JGZZ-ZS0102	ZD3H9FF7GAZZ-ZS0102
+ 15 KV	B	B	B
- 15 KV	B	B	B

- ESD showed pass result.

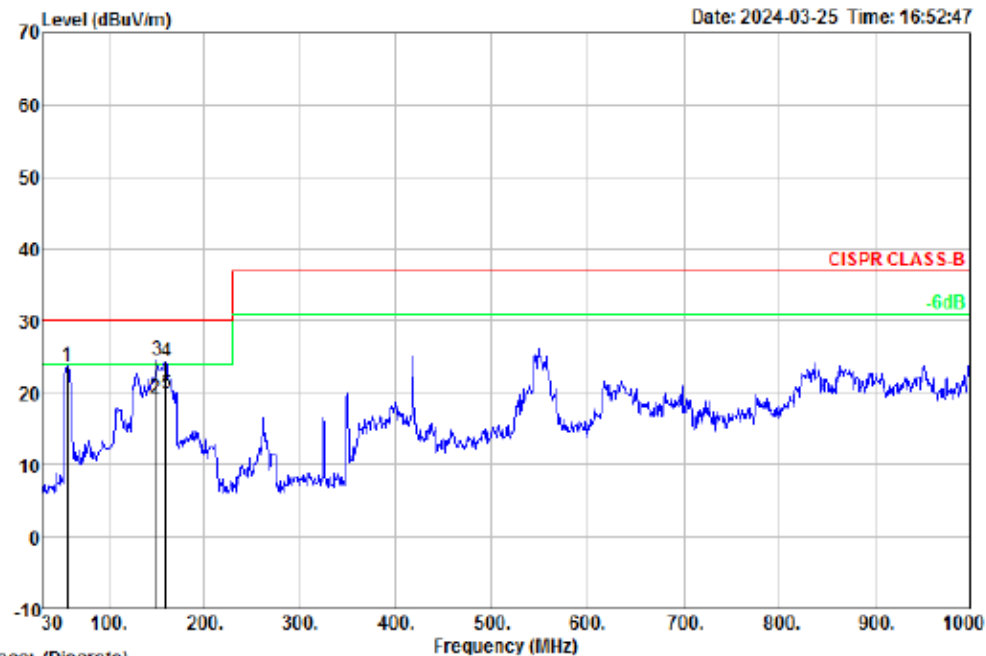
ESD Test Report Form

☐ Record ☒ ESD check ☐ Customer check

File Name	20240321-5D2430032-G230HAN01.1			Test Ambient:	22 °C / 51% RH						
Application Sec.	RRCAB4	User	黃泓儒	Ext.	86-503004	Sample Received	2024/3/21				
Model Name	G230HAN01.1	Quantity (pcs)	3	Panel S/N	ZD3H9FF7NEZZ-ZS0102	(#1)					
					ZD3H9FF7JGZZ-ZS0102	(#2)					
					ZD3H9FF7GAZZ-ZS0102	(#3)					
Examiner	王溪源	Ext.	504202	Date Tested	2024/3/22	Date Completed	2024/3/22				
Test standard	☒ AUO Fixed Point Position Rules ☐ AUO IC Nearly Point Position Rules ☐ Customer Rules										
Stage	TEST		Resolution	Frame rate (Hz)	Color depth (bit)	Pixel clock (MHz)	Level B				
			1920 X 1080								
XPCB Part №		Layout №		PCB Vendor	N/A						
YPCB Part №		Layout №		PCBA Vendor	N/A						
C/B Part №		Layout №		ASIC							
Inverter Part №	NA	Inverter Vendor	NA	SOURCE IC							
				GATE IC							
Purposes	其他,變更TCOON BD驗證										
Test Specification: Test Level Requirement: B					Results PASS						
Test Standard: AUO Fixed Point Position Rules (8 points, 25 times/point)											
Acceptance Definition:											
Class A		Normal operation. No degradation. No failures.									
Class B		Some performance degradation allowed.									
Class C		Temporary performance degradation. Recovery by operator (Reset) is acceptable. No Hardware failures.									
Class D		Permanent damage.									

EMI result

- EMI showed pass result. (Horizontal)



Trace: (Discrete)

Site : L3C Chamber

Condition: CISPR CLASS-B 10m 24(2020) HORIZONTAL

Model : G230HAN01.1

Mode : 1920*1080*60

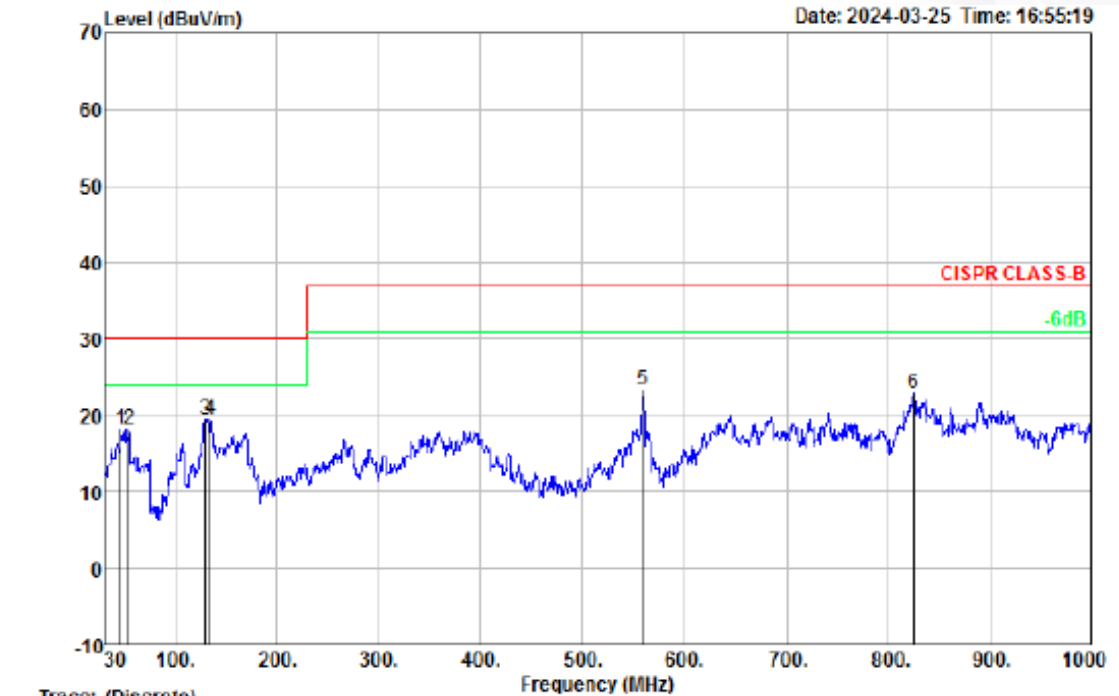
pattern : patternGen Hs

: SS 3%

	Freq	Level	Read	Limit	Over	Remark
	MHz	dBuV/m	Level	Line	Limit	
	MHz	dBuV/m	dBuV	dB/m	dBuV/m	dB
1	57.16	23.69	23.53	0.16	30.00	-6.31 Peak
2	150.01	19.17	20.00	-0.83	30.00	-10.83 QP
3 p	150.28	24.60	25.43	-0.83	30.00	-5.40 Peak
4 !	159.98	24.30	25.02	-0.72	30.00	-5.70 Peak
5 q	159.98	19.78	20.50	-0.72	30.00	-10.22 QP

EMI result

- EMI showed pass result. (Vertical)



Trace: (Discrete)

Site : L3C Chamber

Condition: CISPR CLASS-B 10m 24(2020) VERTICAL

Model : G230HAN01.1

Mode : 1920*1080*60

pattern : patternGen He

: SS 3%

	Freq	Level	Read	Limit	Over	Remark
	MHz	dBuV/m	Level	Line	Limit	
			dBuV	dB/m	dBuV/m	dB
1	46.49	18.26	14.15	4.11	30.00	-11.74 Peak
2	54.25	18.22	17.68	0.54	30.00	-11.78 Peak
3 p	129.91	19.54	15.64	3.90	30.00	-10.46 Peak
4	134.76	19.42	14.40	5.02	30.00	-10.58 Peak
5	560.59	23.28	12.60	10.68	37.00	-13.72 Peak
6	825.40	22.86	7.48	15.38	37.00	-14.14 Peak



Tap Into The Possibilities

